AI FOR GOOD 2.0 · BLOOD WARRIORS

PulseLink

The Blood Subscription for Thalassemia Care

Blood for a Thalassemia patient should not be hunted in a panic every two weeks. It should arrive — on schedule, from donors the system already trusts. PulseLink makes recurring blood supply behave like a subscription the patient never has to think about.

Team · Gafoor & Misbah

Powered by a Donor Reliability Network

Theme: AI for Blood Donation & Care Coordination | Problem: Optimizing donor–patient matching using AI

THE OPERATIONAL PROBLEM

A lifelong, predictable need — served by a system with no memory

A child with Thalassemia Major needs blood every 2–3 weeks, for life. The schedule is known. Yet every cycle, parents and volunteers start from zero — broadcasting SOS messages to strangers. The blood system treats each request as a first-time emergency. It never remembers who helped last time.

1,00,000+

Thalassemia Major patients in India on lifelong recurring transfusions

Every 2–3 wks

A known, repeating cycle — the most predictable demand in the blood system

Re-sourced ×26

Same patient, same blood group — sourced from scratch every cycle, every year

0 memory

Donors are anonymous one-offs; the system never learns who is reliable

The deeper failure: it is not just that requests are last-minute — it is that the system has no idea which donors actually show up. Reliability is invisible, so it cannot be used.

WHY CURRENT TOOLS DON'T WIN THIS

Everyone built a faster way to ask. Nobody built memory.

SOS broadcast apps

Blast every nearby donor. Reaches many, commits none — and treats a proven donor and a stranger exactly the same.

Donor directory apps

A searchable contact list. The parent is still the dispatcher, and the list says nothing about who is dependable.

Generic AI chatbots

Answer awareness FAQs. Useful — but they do not move a unit of blood, and they learn nothing about donor behaviour.

The blind spot

Every tool treats donors as interchangeable and forgets them after one ask. None of them build the one asset that would fix this:

a memory of who is reliable — and the foresight to commit them early.

PulseLink's move: give the blood system a memory — then use it to deliver blood on a schedule, not on a prayer.

THE BIG IDEA

Blood should not be hunted. It should arrive.

PulseLink turns each patient's recurring transfusion into a Blood Subscription: a standing supply line of pre-committed, rotating donors that renews itself automatically — for life. The patient stops making requests. Blood simply shows up on time.

Today · Hunting

- → Parent realises blood is due in 2 days
- → Panic broadcast to strangers on WhatsApp
- → Begging, calling, tracking who said yes
- → Repeat from zero — every 2 weeks, forever

With PulseLink · Subscription

- → Next 3 transfusions forecast in advance
- → Trusted donors auto-rostered into each slot
- → Donors confirm a calendar slot, not an SOS
- → Subscription renews itself — patient does nothing

The catch: a subscription only works if the donors are dependable. That is what the engine on the next slide guarantees.

THE ENGINE — OUR CORE INNOVATION

The Donor Reliability Network: giving blood a memory

PulseLink builds a living graph of every donor. Each donation outcome — showed up, declined, ghosted, with how much notice, in which season — updates a Reliability Score. The network learns who to depend on, and routes each slot accordingly.

What the Reliability Score learns from

- **Show-up history**
  Did past confirmations turn into actual donations?
- **Notice sensitivity**
  How much lead time this donor needs to say yes
- **Seasonal pattern**
  Festivals, exams, monsoon — when they go quiet
- **Recency & fatigue**
  Time since last ask; how often they've been tapped
- **Response speed**
  How fast they reply — a signal for time-critical slots

How the network routes a slot

Anchor donors

High score — take the time-critical, near-term slots

Steady donors

Reliable with notice — fill the planned, routine slots

Growing donors

New or shaky — given low-stakes asks to build trust

The network grows its own future Anchor donors over time.

END-TO-END FLOW

From a messy record to a subscription that renews itself

1 · Ingest

An LLM reads messy real records — WhatsApp notes, registers — into clean structured patient + donor data.

2 · Forecast

Engineered time features predict each patient's next transfusion dates with a confidence window.

3 · Route

The Reliability Network assigns each slot to the right tier of donor — anchor, steady or growing.

4 · Commit

Donors get a scheduled slot in their language; an LLM tailors the message. They tap Accept.

5 · Renew

After each donation the score updates and the next cycle is auto-rostered. The subscription self-heals.

What each person actually sees

Parent: a calm home screen — “Next transfusion 14 Jun · Donor confirmed · Backup ready.” Donor: a scheduled slot with a fair, friendly ask. Coordinator: one dashboard of subscriptions, sorted by risk, so they act only on the few that need a human.

Four things that make this hard to copy

Reliability as infrastructure

The blood system has no trust layer. PulseLink builds one — a donor graph that learns and compounds in value over time.

Engineered time features

Not a naive average. Interval-trend, seasonal donation-friction and Hb-decay slope sharpen every forecast.

LLMs used invisibly

No chatbot. LLMs parse messy real-world records into clean data and tailor each donor message by language and tone.

Self-healing subscriptions

Every slot has a ranked backup chain. A decline silently promotes the next donor — the subscription degrades gracefully.

Why this scores on originality: the novelty is not AI or an app — it is treating donor reliability as a learnable asset that makes a lifelong subscription possible.

SYSTEM ARCHITECTURE

How data flows — messy record to confirmed donor

DATA & PROCESSING PIPELINE

STEP 1
Raw Intake
WhatsApp note, register, form

→

STEP 2
LLM Extraction Service
Parses free text → structured records

→

STEP 3
Feature Pipeline
Interval-trend, seasonal, Hb-decay

→

STEP 4
Forecast + Reliability Engine
Due-date model + donor scoring

→

STEP 5
Tiered Routing Engine
Assigns slot → anchor / steady / growing

→

STEP 6
Delivery Service
Scheduled, localised donor request

CONSUMES THE PIPELINE OUTPUT

Patient App
Subscription status & next transfusion

Donor App
Accept / decline a scheduled slot

Coordinator Dashboard
Risk-sorted view of all subscriptions

</> FOUNDATION · shared across every layer

API layer
Python / FastAPI — REST services

Datastore
PostgreSQL — patients, donors, donor graph

Frontend
React — web + responsive mobile

Messaging
SMS / WhatsApp / push gateway

Safeguards
Consent, role-based access, audit trail

IMPACT

Who benefits, and how much

Patients & families

Blood arrives on time. Parents stop being unpaid dispatchers and get a status they can build life around.

Donors

A fair, scheduled ask they can keep. The network grows them from first-timers into trusted anchors — they stay for years.

Blood Warriors & volunteers

Coordinators manage subscriptions by risk on one dashboard, instead of firefighting every case by hand.

Partner blood banks

Predictable, smoothed demand and a shared reliability signal make network-wide inventory planning possible.

On-time

transfusions become the default, not the exception

Trust compounds

the donor network gets more reliable every cycle

Manage by risk

one dashboard replaces dozens of WhatsApp threads

FEASIBILITY & SCALABILITY

What we ship at the hackathon — and where it goes

Hackathon prototype (the demo)

- ✓ LLM parses a messy sample record into clean data — live
- ✓ Forecast view: next transfusion dates per patient
- ✓ Reliability Network with scored, tiered donors
- ✓ Auto-rostered Blood Subscription per patient
- ✓ Donor screen: accept / decline a scheduled slot
- ✓ Decline → backup auto-promoted; score updates live

Path to scale (after the hackathon)

Across users

Onboard real Blood Warriors patients and donors, city by city.

Across cities

Each city is a new donor graph — the engine is location-agnostic.

Across languages

LLM messaging localises to each donor's regional language.

Across partners

A shared reliability signal other NGOs and blood banks can plug into.

Realistic by design: the prototype runs on seeded data, so it fully demos without depending on live integrations to prove the idea.

HONEST RISKS & ASSUMPTIONS

What could go wrong — and how we've planned for it

Cold start — no reliability data yet

Mitigation: New donors begin at a neutral score and are grown via low-stakes asks; broadcast mode covers gaps until the network matures.

Transfusion dates aren't perfectly regular

Mitigation: The forecast gives a window, not a fixed day, and re-learns after every actual transfusion; a coordinator confirms before locking.

Thin donor pools in small towns

Mitigation: PulseLink flags low-coverage subscriptions early and falls back to broadcast — it never blocks an emergency path.

Sensitive medical & donor data

Mitigation: Data minimisation, role-based access, explicit consent and an audit trail — detailed on the next slide.

Key assumptions: patients/NGOs can share basic transfusion history; a willing donor base exists per region; donors carry a phone with SMS or WhatsApp.

Handling blood, health data and trust with care

Data minimisation

Only what a match needs — blood group, area, eligibility. No diagnoses or clinical records beyond transfusion dates.

Consent & control

Donors opt in, set their own frequency limits, and can pause or leave anytime. Patients consent to subscription sharing.

Role-based access

Patients, donors and coordinators see only their own data. No open directory of either group.

AI kept advisory

LLMs only parse records and draft messages; the network only suggests. A human coordinator confirms every critical step.

Fair reliability scoring

Scores can only raise a donor's standing, never punish or expose them. Low scores are never shown to other users.

Accessibility

Works on low-end phones via SMS, in regional languages, and never forces an app install on a patient in crisis.

THE TAKEAWAY

Stop hunting for blood. Subscribe to it.

Thalassemia gives us a need we can see coming. PulseLink turns that foresight into a Blood Subscription — and builds a donor network that gets more reliable every single cycle. No child should wait on blood that could have been ready weeks ago.

Subscribe

blood arrives on schedule

Trust

a donor network with memory

Renew

it self-heals, every cycle

PulseLink · Team Gafoor & Misbah